

11 — Probabilistic models 1

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Machine Learning (MPA-MLR)

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1 • Probability introduction

Distributions, joint, marginal and conditional probability, Bayes' rule.

2 • Model parameters estimation

A model is a distribution; its parameters have a distribution too.

3 • Maximum likelihood (MLE)

The parameters under which the observed data are the most probable.

4 • Maximum a-posteriori (MAP)

The most probable parameters given the data — likelihood plus a prior.

5 • Bayesian inference

Not one estimate but the whole distribution of the parameters.

Next lecture

Probabilistic models 2 — naive Bayes, Gaussian mixture models with EM, logistic regression.

1 · Probability introduction

Everything today is built from four objects: a distribution, a joint, a marginal and a conditional.

Probability — a measure quantifying the likelihood that an event will occur. The probability of an event A is written as

$$P(A) \quad (\text{or } p(A) \text{ or } \Pr(A))$$

Random variable X — a variable whose value is uncertain: a dice roll, a covid test result, a class label in a dataset. The probability that X takes the value x_1 is $P(X = x_1)$.

Probability distribution — the probabilities of occurrence of all possible outcomes of a random variable, i.e. the values $P(X = x_i)$ for all x_i .

Notation for the whole course: capital $P(\cdot)$ for the probability of an event or of a discrete value, lower-case $p(\cdot)$ for a density and for the short form $p(x) = P(X = x)$. The third lecture already used $p(x)$, $p(x, y)$ and $p(x \mid y)$ in the entropy formulas — it is the same notation.

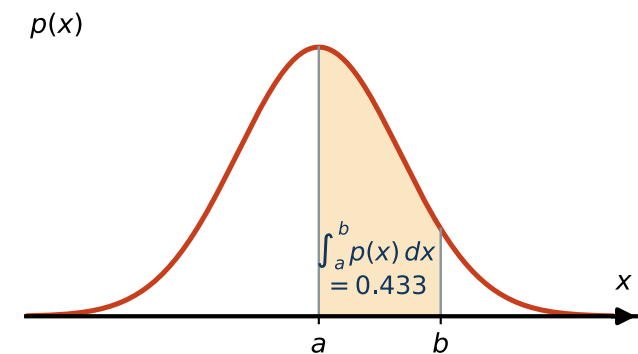
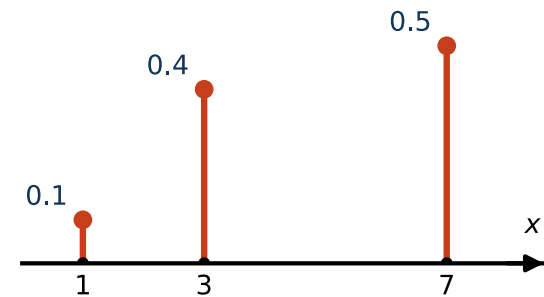
Probability mass function (PMF) — discrete variable:

$$P(X = x) = p(x), \quad \sum_x p(x) = 1$$

Probability density function (PDF) — continuous variable:

$$P(X = x) = 0, \quad P(x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} p(x) dx$$
$$\int_{-\infty}^{\infty} p(x) dx = 1$$

For a continuous variable, only an **interval** has a non-zero probability; the density itself is not a probability and may exceed 1.



Non-parametric description — only for a **discrete** distribution; it is given directly by the probability of each value.

$$x \in \{1, 3, 7\}$$

$$P(X = 1) = 0.1$$

$$P(X = 3) = 0.4$$

$$P(X = 7) = 0.5$$

$$x \in \{0.1, 1.5, 2.3, 26.7\}$$

$$P(X = 0.1) = 0.10$$

$$P(X = 1.5) = 0.25$$

$$P(X = 2.3) = 0.38$$

$$P(X = 26.7) = 0.27$$

$$x \in \{0, 1\}, y \in \{0, 1\} \text{ — joint:}$$

$$P(X = 0, Y = 0) = 0.1$$

$$P(X = 1, Y = 0) = 0.2$$

$$P(X = 0, Y = 1) = 0.4$$

$$P(X = 1, Y = 1) = 0.3$$

Each of the three sets sums to 1 — that is the only condition a non-parametric description has to satisfy.

Non-parametric distribution from observed data

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Feature table — one row per subject:

Subject	Feature 1	Decision
01	Blue	Positive
02	Blue	Negative
03	Brown	Positive
...	...	...
N	Green	Positive

Contingency table — counts:

	Blue	Green	Brown	Total
Positive	53	47	45	145
Negative	65	33	82	180
Total	118	80	127	325

Joint distribution — divide by 325:

	Blue	Green	Brown
Positive	0.163	0.145	0.138
Negative	0.200	0.102	0.252

Marginals (sum over the other variable):

$$p(\text{Positive}) = \frac{145}{325} = 0.446,$$

$$p(\text{Blue}) = \frac{118}{325} = 0.363$$

From the feature table we can build any contingency table, but not the other way around. What happens if we have 3 features? Can we build a contingency table from a *continuous* variable?

Parametric description — the shape of the distribution is given by a formula with a few parameters. It works for continuous **and** discrete variables.

E.g. the 1D normal distribution is defined by a mean μ and a standard deviation σ :

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

In general we have a **vector of parameters** θ ; for the 1D normal distribution $\theta = [\mu, \sigma^2]$ — throughout this lecture the normal distribution is written through the **variance** σ^2 , not through σ .

The whole rest of the lecture is about one question: **given data, which θ do we choose?**

Example — Bernoulli distribution

Parametric

$$f(x \mid \theta) = \theta^x (1 - \theta)^{1-x}$$

$$x \in \{0, 1\}, \quad \theta = 0.4$$

Non-parametric

$$P(X = 0) = 0.6, \quad P(X = 1) = 0.4$$

From observed data of a **discrete** variable we can obtain both descriptions. For a **continuous** variable only the parametric one exists.

Feature table

Subject	Feature 1
01	1
02	0
03	0
...	...
N	1

The Bernoulli variable is the one behind DropOut in the sixth lecture, and it will be the likelihood of logistic regression in the next lecture. Here θ is a **scalar** parameter — the probability of a 1.

Joint probability — the probability of two events happening together. **Joint probability distribution** — the joint probability function of two or more random variables.

Discrete case (mass function)

$$P(X = x, Y = y) = P(X = x \text{ and } Y = y) = p(x, y)$$

Continuous case (density function)

$$P(x_1 \leq X \leq x_2, y_1 \leq Y \leq y_2), \quad p(x, y)$$

Independent variables

$$p(x, y) = p(x) \cdot p(y)$$

Dependent variables — chain rule

$$p(x, y) = p(x \mid y) \cdot p(y)$$

Marginal probability is the **sum of a joint distribution along one or more dimensions**. The summed-out variables are *marginalized out*; the process is *marginalization*.

Marginalizing along Y :

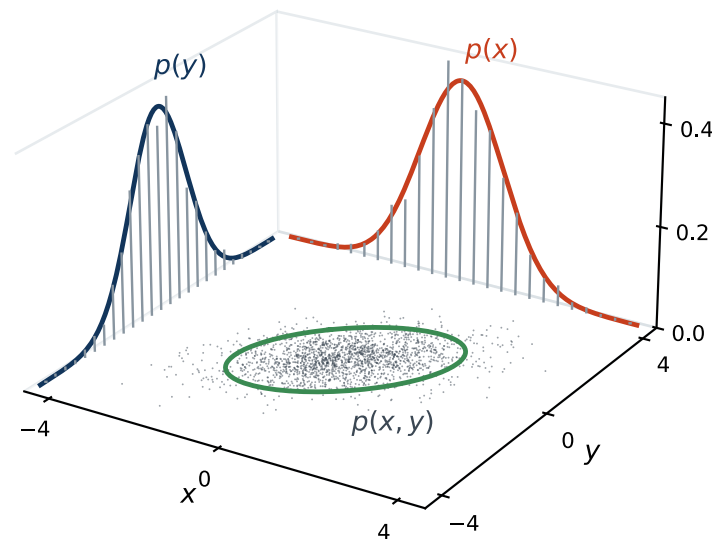
$$P(X = x) = \sum_y P(X = x, Y = y)$$

Marginalizing along X :

$$P(Y = y) = \sum_x P(X = x, Y = y)$$

Abbreviated notation — **law of total probability**:

$$p(x) = \sum_y p(x, y)$$



Conditional probability — the probability that an event A occurs given that another event B has occurred, written $P(A \mid B)$ (or $P(X \mid Y = y)$ for random variables). It is one of the most fundamental concepts in probability theory.

Kolmogorov definition: the conditional probability of A given B is the ratio of the probability that both A and B occur to the marginal probability of B :

$$p(x \mid y) = \frac{p(x, y)}{p(y)} \quad \Longleftrightarrow \quad p(x, y) = p(x \mid y) \cdot p(y)$$

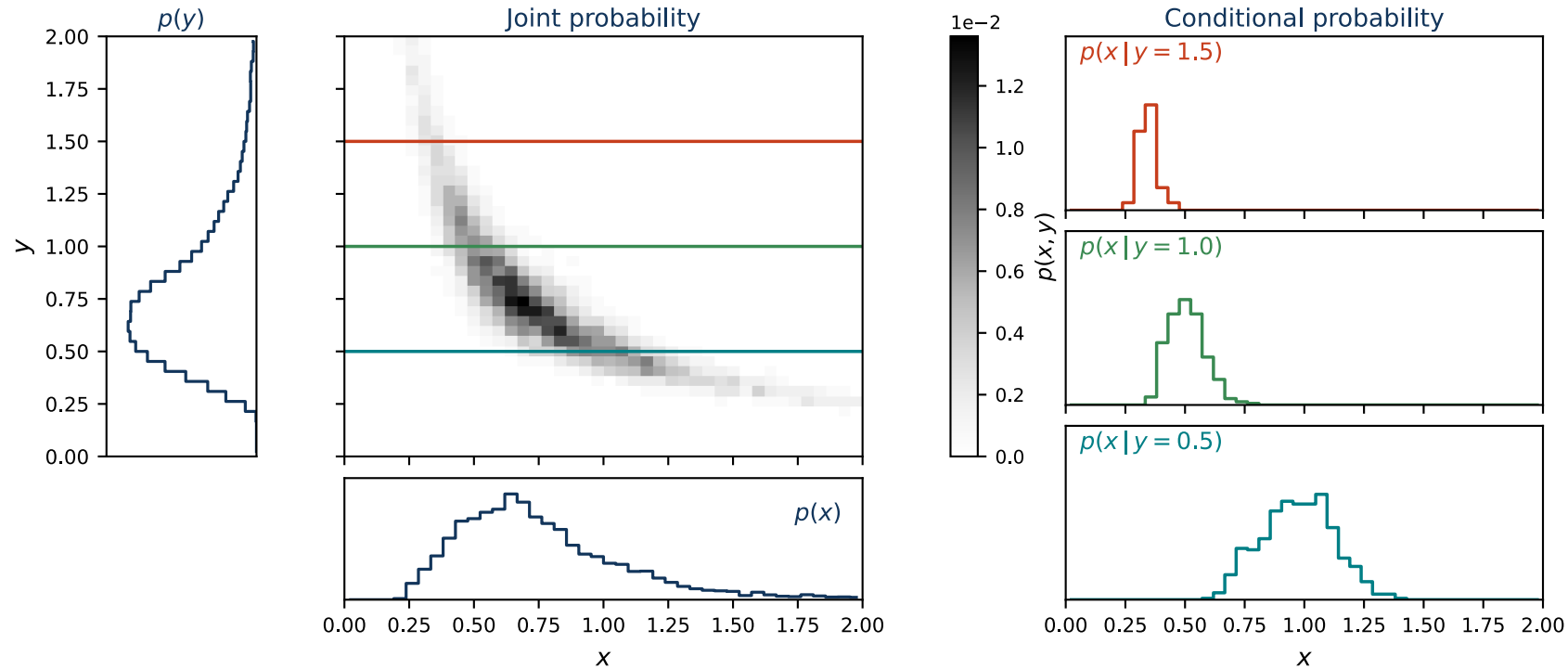
In contrast with the joint probability, the conditional probability is **not symmetric** — in general

$$p(x \mid y) \neq p(y \mid x)$$

By hand, from the contingency table three slides back: $p(\text{Positive} \mid \text{Blue}) = \frac{53}{118} = 0.449$, but $p(\text{Blue} \mid \text{Positive}) = \frac{53}{145} = 0.366$ — the same joint count 53 divided by two different marginals.

Joint, conditional and marginal probability — example

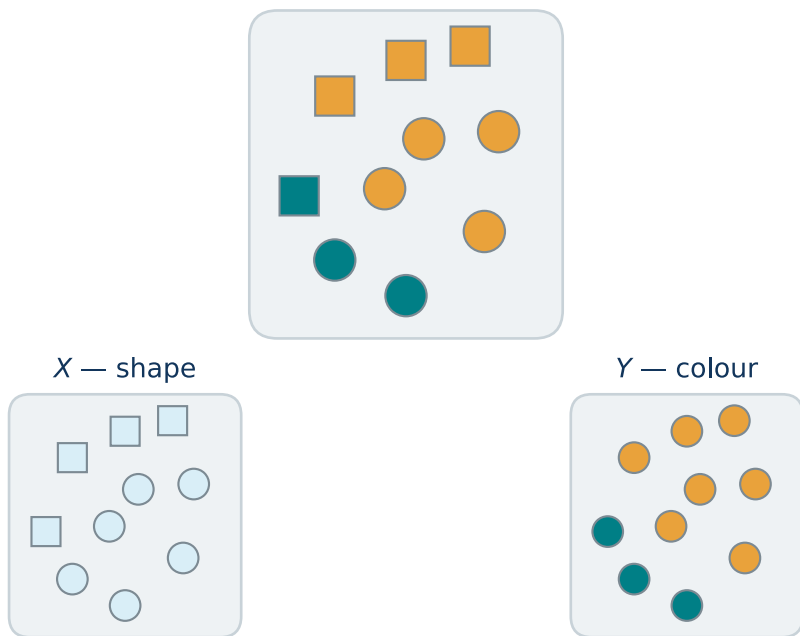
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Redrawn after Ivezić, Connolly, VanderPlas & Gray, *Statistics, Data Mining, and Machine Learning in Astronomy* (astroML, [fig_conditional_probability](#)).

Probability distribution example — shape and colour

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X — shape: c (circle), s (square); Y — colour: b (blue), o (orange).

Counting the 10 objects (4 squares, 1 of them blue; 6 circles, 2 of them blue):

$$p(s) = \frac{4}{10} = 0.4, \quad p(c) = 0.6, \quad p(b) = \frac{3}{10} = 0.3, \quad p(o) = 0.7$$

$$p(b, s) = \frac{1}{10} = 0.1 = p(s, b)$$

$$p(b | s) = \frac{p(b, s)}{p(s)} = \frac{0.1}{0.4} = 0.25 \quad p(s | b) = \frac{p(s, b)}{p(b)} = \frac{0.1}{0.3} = 0.33$$

$$p(y) = \sum_x p(x, y) = \sum_i p(y | x_i) p(x_i)$$

From the definition of conditional probability:

$$p(x, y) = p(x \mid y) p(y) = p(y, x) = p(y \mid x) p(x) \implies p(x \mid y) p(y) = p(y \mid x) p(x)$$

Bayes' rule:

$$\underbrace{p(x \mid y)}_{\text{posterior}} = \frac{\overbrace{p(y \mid x)}^{\text{likelihood}} \overbrace{p(x)}^{\text{prior}}}{\underbrace{p(y)}_{\text{evidence}}} = \frac{p(y \mid x) p(x)}{\sum_i p(x_i, y)} = \boxed{\frac{p(y \mid x) p(x)}{\sum_i p(y \mid x_i) p(x_i)}}$$

The denominator is just the law of total probability (the sum runs over all values x_i that X can take) — it makes the posterior sum (or integrate) to 1.

Bayes rule example 1 — smoking and heart attack

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A ... a person has a heart attack

B_1 ... non-smoker, B_2 ... light smoker, B_3 ... heavy smoker

Measurement (100 people with a heart attack):

	B_1	B_2	B_3	Total
A	50	30	20	100

$$\Rightarrow p(B_1 | A) = 0.5, p(B_2 | A) = 0.3, p(B_3 | A) = 0.2$$

$$\text{Prior: } p(B_1) = 0.9, p(B_2) = 0.07, p(B_3) = 0.03, \\ p(A) = 0.05$$

Posterior:

$$p(A | B_1) = \frac{p(B_1 | A) p(A)}{p(B_1)} = \frac{0.5 \cdot 0.05}{0.9} = 0.0278$$

$$p(A | B_2) = \frac{0.3 \cdot 0.05}{0.07} = 0.2143$$

$$p(A | B_3) = \frac{0.2 \cdot 0.05}{0.03} = 0.3333$$

Check with the law of total probability:

$$\sum_i p(A | B_i) p(B_i) = 0.0278 \cdot 0.9 + 0.2143 \cdot 0.07 + 0.3333 \cdot 0.03 \\ = 0.05 = p(A)$$

A heavy smoker has a **12× higher** risk than a non-smoker — and the measurement alone (50 : 30 : 20) says nothing of the sort. Only the prior turns it into a risk.

Bayes rule example 2 — a positive test

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A ... a disease occurs B_+ ... a positive test with **sensitivity and specificity of 98 %**

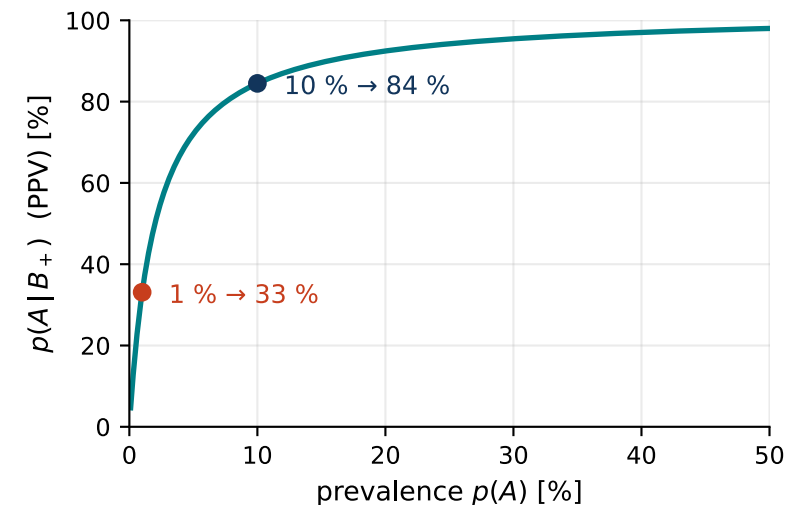
Case 1: I have a positive test. What is the probability that I really have the disease?

Prevalence 10 %

$$\begin{aligned} p(B_+) &= p(B_+ | A) p(A) + p(B_+ | \bar{A}) p(\bar{A}) \\ &= 0.98 \cdot 0.1 + 0.02 \cdot 0.9 = 0.116 \\ p(A | B_+) &= \frac{0.98 \cdot 0.1}{0.116} = 0.845 \approx \mathbf{84\%} \end{aligned}$$

Prevalence 1 %

$$\begin{aligned} p(B_+) &= 0.98 \cdot 0.01 + 0.02 \cdot 0.99 = 0.0296 \\ p(A | B_+) &= \frac{0.98 \cdot 0.01}{0.0296} = 0.331 \approx \mathbf{33\%} \end{aligned}$$



$p(A | B_+)$ is exactly the **PPV (precision)** from the second lecture; sensitivity is $p(B_+ | A)$ and specificity is $p(B_- | \bar{A})$. [code example — Bayes and prevalence](#)

2 · Model parameters estimation

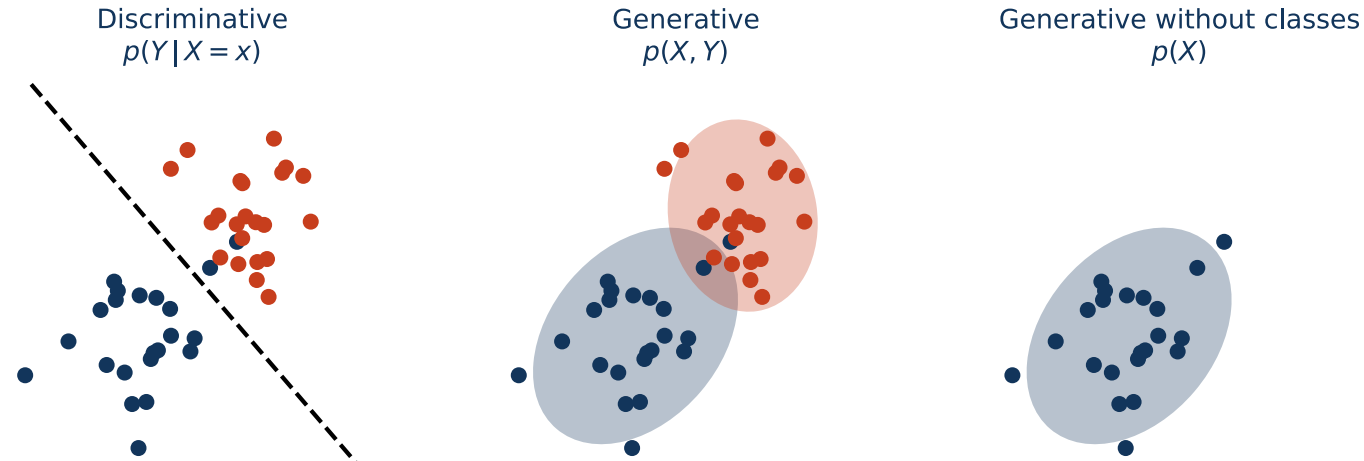
The model is a distribution. Its parameters are uncertain too — so they have a distribution as well.

Probabilistic models incorporate random variables and probability distributions into the model of an event or phenomenon. While a deterministic model gives a single possible outcome, a **probabilistic model** gives a **probability distribution** as a solution.

- We want a model of a probability distribution $p(x)$.
- It can be specified parametrically, with parameters θ : $p_\theta(x)$.
- Given a measured dataset D , the parameters themselves have a distribution:

$$p(\theta \mid D) = p(\theta \mid x_1, x_2, \dots, x_n)$$

- If we know $p(\theta \mid D)$, we can pick the most probable parameters and use them in the model $p_\theta(x)$.



Discriminative — $f : X \rightarrow Y$, models $p(Y | X = x)$

SVM, k -NN, logistic regression

Generative — models $p(X, Y)$

naive Bayes classifier, conditional GANs, text generation with transformers

Generative without classes — models $p(X)$

Gaussian fitted with MLE or MAP, GANs

Prior probability $p(\boldsymbol{\theta})$ — the distribution of the parameters before any data is observed (what we know before the experiment).

Likelihood $p(D \mid \boldsymbol{\theta})$ — the distribution of the observed data given the parameters. It quantifies how likely a model with parameters $\boldsymbol{\theta}$ is to generate the data D .

Posterior probability $p(\boldsymbol{\theta} \mid D)$ — the distribution of the parameters after taking the observed data into account.

Marginal likelihood (evidence) $p(D)$ — the distribution of the observed data marginalized over the parameters:

$$p(D) = \sum_i p(D \mid \boldsymbol{\theta}_i) p(\boldsymbol{\theta}_i) \quad \text{or} \quad p(D) = \int_{\Theta} p(D \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) d\boldsymbol{\theta}$$

Data D and model parameters $\boldsymbol{\theta}$ — the same Bayes' rule as before, only $x \rightarrow \boldsymbol{\theta}$ and $y \rightarrow D$:

$$\underbrace{p(\boldsymbol{\theta} \mid D)}_{\text{posterior}} = \frac{\overbrace{p(D \mid \boldsymbol{\theta})}^{\text{likelihood}} \cdot \overbrace{p(\boldsymbol{\theta})}^{\text{prior}}}{\underbrace{p(D)}_{\text{marginal likelihood}}}$$

The likelihood is **not** a distribution of $\boldsymbol{\theta}$ — it does not integrate to 1 over $\boldsymbol{\theta}$. It is a distribution of the data, read as a function of the parameters.

Example — model parameters estimation

We have n measurements of human height $D = \{x_1, x_2, \dots, x_n\}$, assumed $x_i \sim \mathcal{N}(\mu, \sigma^2)$:

$$p(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

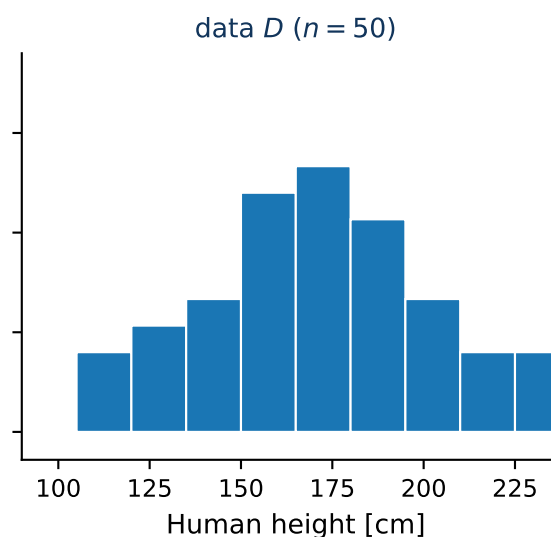
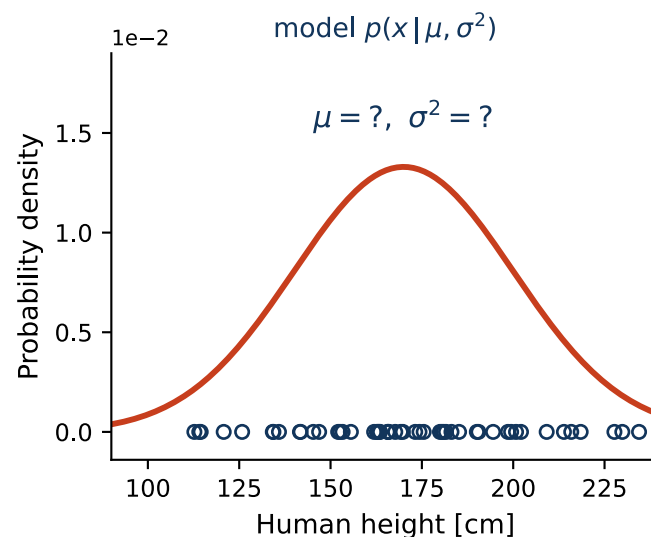
$x_1 = 172$ cm

$x_2 = 189$ cm

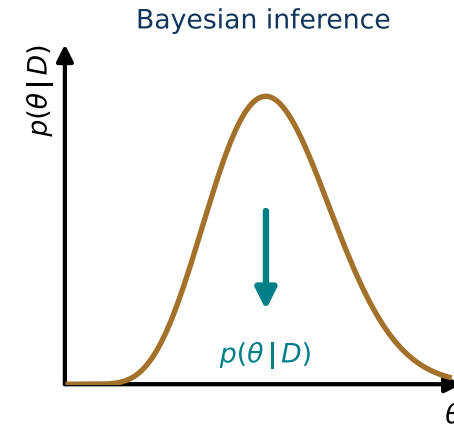
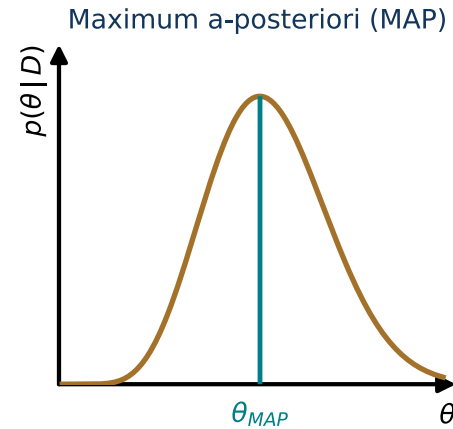
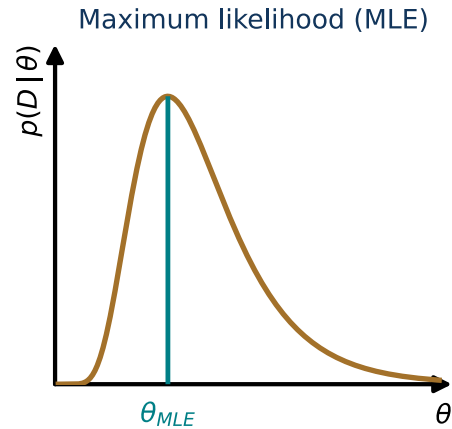
$x_3 = 168$ cm

$\vdots$

$x_n = 163$ cm



We are interested in the distribution of the model parameters $p(\boldsymbol{\theta} \mid D)$, and in the most probable parameters $\arg \max_{\boldsymbol{\theta}} p(\boldsymbol{\theta} \mid D)$, where $\boldsymbol{\theta} = [\mu, \sigma^2]$.



Maximum likelihood (MLE) — under the assumed model the observed data are the most probable.

$$\arg \max_{\theta} p(D | \theta)$$

Maximum a-posteriori (MAP) — the most probable model parameters for the given data.

$$\begin{aligned} \arg \max_{\theta} p(\theta | D) &= \arg \max_{\theta} \frac{p(D | \theta) p(\theta)}{p(D)} \\ &= \arg \max_{\theta} p(D | \theta) p(\theta) \end{aligned}$$

$p(D)$ is a constant, so it can be dropped inside an $\arg \max$.

Bayesian inference — the whole distribution of the parameters for the given data.

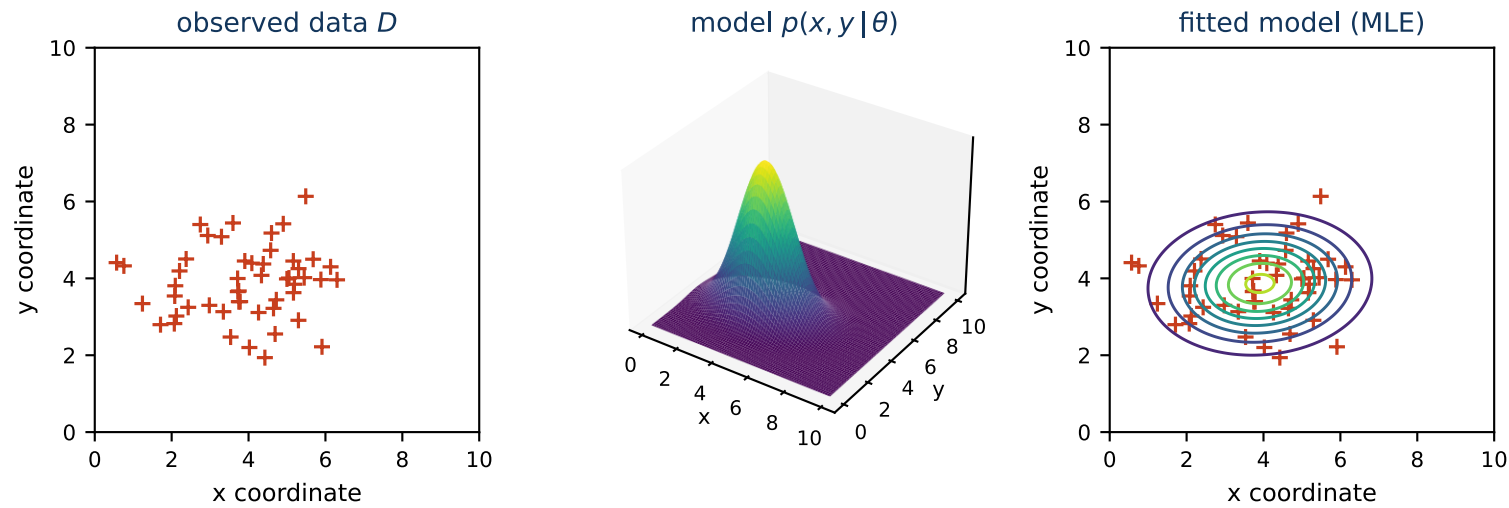
$$\begin{aligned} p(\theta | D) &= \frac{p(D | \theta) p(\theta)}{p(D)} \\ &= \frac{p(D | \theta) p(\theta)}{\int_{\Theta} p(D | \theta) p(\theta) d\theta} \end{aligned}$$

The integral is hard to compute.

3 · Maximum likelihood

Choose the parameters that make the data you actually saw as probable as possible.

The estimation method that finds the parameters of a probability distribution by maximizing a likelihood function for the set of observed data $D = \{x_1, x_2, \dots, x_n\}$. The resulting model is the most likely model for generating the observed data.



The main advantage is that many problems have an **analytic solution**, so the parameters of a preselected distribution are easy to find. It can also be solved by any optimization technique (brute force, iterative algorithms).

Task: find the θ that maximizes the likelihood

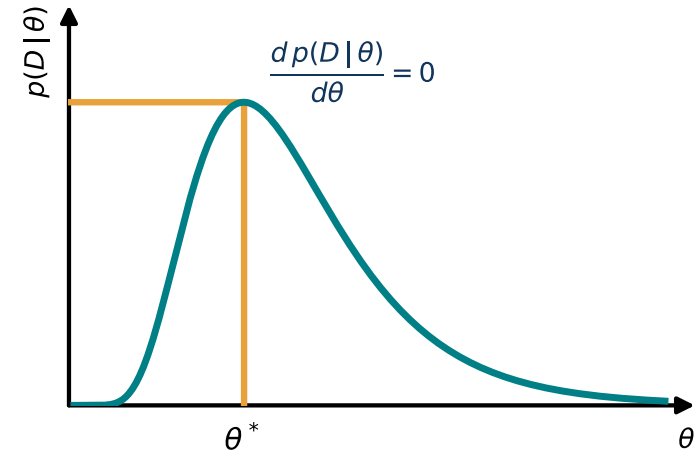
$$\theta^* = \arg \max_{\theta} p(D \mid \theta)$$

If we know that $x_1, x_2, \dots, x_n \sim \mathcal{N}(\theta)$ with $\theta = [\mu, \sigma^2]$, then

$$\mu^* = \arg \max_{\mu} p(D \mid \mu, \sigma^2)$$

$$\sigma^{2*} = \arg \max_{\sigma^2} p(D \mid \mu, \sigma^2)$$

The maximum is where the derivative vanishes — for a smooth likelihood this gives a system of equations, one per parameter.

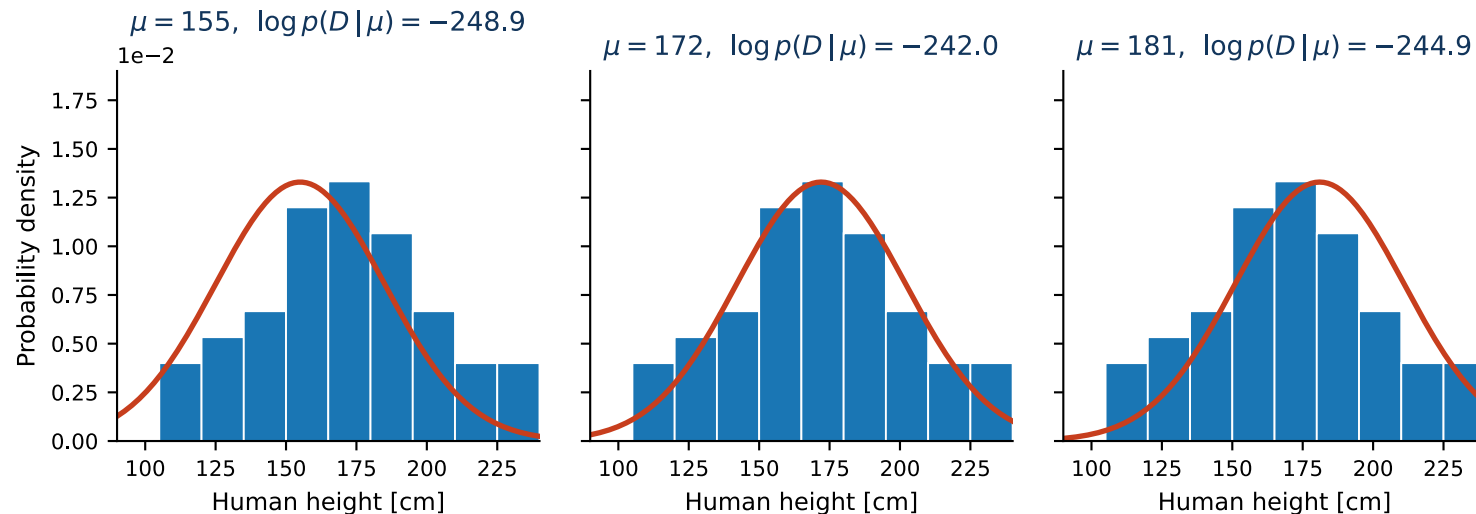


PDF of the normal distribution:

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

For n independent identically distributed (i.i.d.) samples $x_i \sim \mathcal{N}(\mu, \sigma^2)$ the likelihood is a **product**:

$$p(D \mid \mu, \sigma^2) = p(x_1, \dots, x_n \mid \mu, \sigma^2) = \prod_{i=1}^n p(x_i \mid \mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

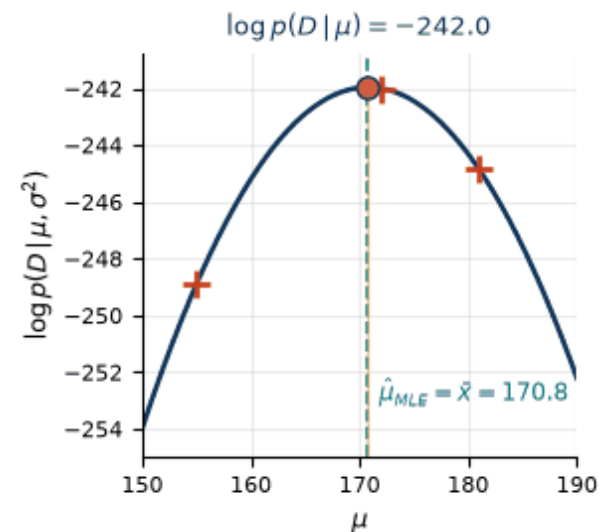
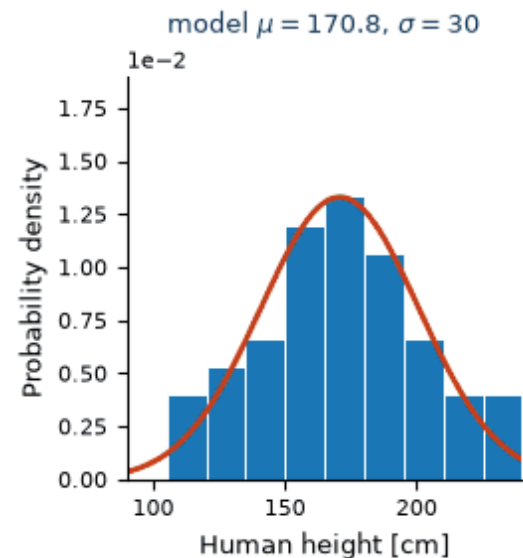


The logarithm turns the product into a sum and does not move the maximum (log is monotone):

$$\ell(\mu) = \log p(D \mid \mu, \sigma^2) = \sum_{i=1}^n \log p(x_i \mid \mu, \sigma^2)$$

For a fixed σ this is a **parabola** in μ , and its vertex is the sample mean.

Three candidate models from the previous slide are three points on that parabola — the best of them is the one closest to the vertex. Sliding μ along the axis moves the model over the data and the point along the parabola.



Start from the likelihood of n i.i.d. samples and take the logarithm:

$$\begin{aligned}\log p(x_1, \dots, x_n \mid \mu, \sigma^2) &= \log \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) \\ &= \sum_{i=1}^n \left[-\log\left(\sqrt{2\pi\sigma^2}\right) - \frac{1}{2\sigma^2}(x_i - \mu)^2 \right] = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\end{aligned}$$

The first term does not contain μ at all, and the second one is (up to a negative constant) the **sum of squared errors**. Maximizing the log-likelihood in μ is therefore the same as minimizing $\sum_i (x_i - \mu)^2$.

That is where the least-squares criterion comes from: it is the maximum likelihood solution under the assumption of additive Gaussian noise.

Derivative with respect to μ , set to zero:

$$\frac{d \log p(x_1, \dots, x_n \mid \mu, \sigma^2)}{d\mu} = \sum_{i=1}^n \frac{1}{\sigma^2} (x_i - \mu) = 0 \Rightarrow \sum_{i=1}^n x_i - n\mu = 0 \Rightarrow \hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

The same procedure for σ^2 gives

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

On our $n = 50$ heights: $\hat{\mu}_{MLE} = 170.79$ cm, $\hat{\sigma}_{MLE}^2 = 934.3$ cm², i.e. $\hat{\sigma}_{MLE} = 30.57$ cm.

By hand — the first three heights only, 172, 189, 168:

$$\hat{\mu}_{MLE} = \frac{172+189+168}{3} = 176.33, \quad \hat{\sigma}_{MLE}^2 = \frac{(-4.33)^2 + 12.67^2 + (-8.33)^2}{3} = \frac{248.67}{3} = 82.9 \quad (\hat{\sigma} = 9.1),$$

whereas the unbiased estimate divides by $n - 1 = 2$: $s^2 = 124.3$.

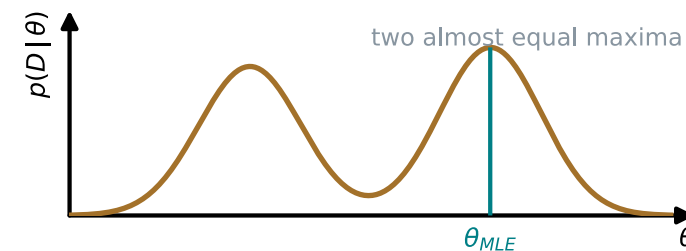
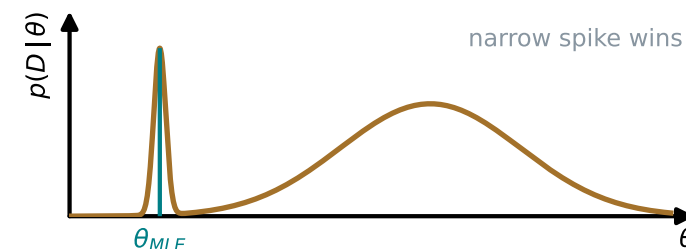
The MLE variance divides by n , not by $n - 1$, so it is **biased**: $E[\hat{\sigma}_{MLE}^2] = \frac{n-1}{n} \sigma^2$. With $n = 50$ it underestimates the variance by 2 % — with $n = 3$ by a third. MLE is not guaranteed to be unbiased.

Pros

- easy to compute — closed-form solution, or a simple maximization problem,
- intuitive interpretation (e.g. the mean for a normal or a Bernoulli distribution).

Cons

- **point estimate** — no representation of uncertainty,
- not suitable for a multi-modal likelihood,
- existence and uniqueness are not guaranteed,
- maybe not what we want,
- **overfitting** — e.g. the “black swan paradox”.



4 · Maximum a-posteriori probability

The same maximization, with one extra factor: what we believed before we saw the data.

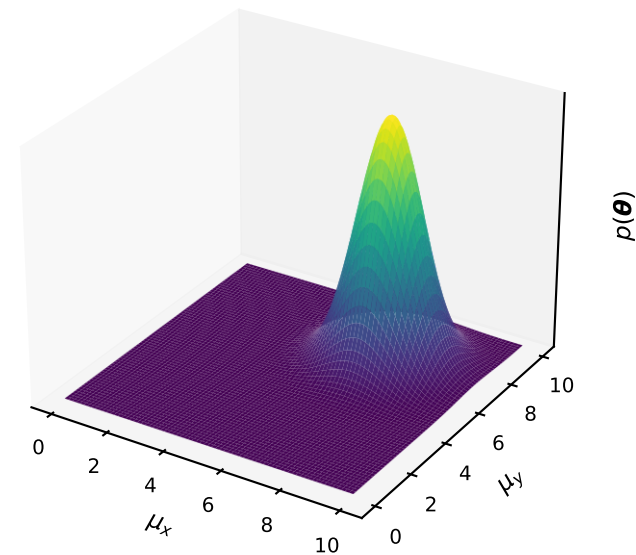
Maximum a-posteriori probability (MAP)

The estimation method that finds the parameters of a probability distribution by maximizing the **a-posteriori probability** $p(\boldsymbol{\theta} \mid D)$ for the observed data

$$D = \{x_1, \dots, x_n\}.$$

The resulting model, defined by the estimated parameters, is **the most probable model for the given (observed) data**, taking into account a prior distribution over the individual models.

So we need one more ingredient than MLE: a distribution $p(\boldsymbol{\theta})$ over the parameters themselves — here over the mean vector $[\mu_x, \mu_y]$ of the two-dimensional model from the previous block.



Maximum likelihood

$$\theta_{MLE}^* = \arg \max_{\theta} p(D \mid \theta)$$

Maximum a-posteriori probability

$$\begin{aligned}\theta_{MAP}^* &= \arg \max_{\theta} p(\theta \mid D) \\ &= \arg \max_{\theta} \frac{p(D \mid \theta) \cdot p(\theta)}{p(D)} \\ &= \arg \max_{\theta} p(x_1, \dots, x_n \mid \theta) \cdot p(\theta)\end{aligned}$$

Two terms may be dropped inside the arg max:

- $p(D)$ — it does not depend on θ , so it is **irrelevant for finding the maximum**;
- $p(\theta)$ — only if we assume a **uniform** prior; then MAP degenerates to MLE.

MAP therefore is

- a **regularization** of the ML estimate,
- a way of using a prior distribution of a related event,
- the **mode** of the posterior $p(\theta \mid D)$, which Bayes' theorem builds from the likelihood $p(D \mid \theta)$ and the prior $p(\theta)$ — still a single point, not the whole posterior.

If the posterior $p(\boldsymbol{\theta} \mid D)$ and the prior $p(\boldsymbol{\theta})$ belong to the same family of distributions, they are called **conjugate distributions** and the prior is a **conjugate prior** for the likelihood $p(D \mid \boldsymbol{\theta})$:

$$p(\boldsymbol{\theta} \mid D) \propto p(D \mid \boldsymbol{\theta}) \cdot p(\boldsymbol{\theta})$$

Examples

- a normal likelihood needs a normal prior (for the mean, with known variance),
- a Bernoulli likelihood needs a beta prior,
- more in the [table of conjugate distributions](#).

Why?

- the analytic solution of the product $p(D \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})$ is much easier than with any other prior,
- the update of the prior by the likelihood becomes transparent — it is just arithmetic on the parameters.

Likelihood of n independent samples, written with the product pulled out:

$$p(x_1, \dots, x_n \mid \mu, \sigma^2) = \prod_{i=1}^n p(x_i \mid \mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp\left(-\frac{\sum_{i=1}^n (x_i - \mu)^2}{2\sigma^2} \right)$$

Prior according to the **conjugate prior** for a normal distribution with known σ of the likelihood, $\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$:

$$p(\mu \mid \mu_0, \sigma_0^2) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left(-\frac{(\mu - \mu_0)^2}{2\sigma_0^2} \right), \quad \mu_0, \sigma_0^2, \sigma^2 - \text{constants}$$

Maximize the logarithm of their product:

$$\mathcal{P}(\mu; \sigma^2, \mu_0, \sigma_0^2) = \log \left[p(x_1, \dots, x_n \mid \mu, \sigma^2) \cdot p(\mu \mid \mu_0, \sigma_0^2) \right]$$

Setting $\frac{d\mathcal{P}}{d\mu} = 0$ gives a **convex combination** of the sample mean and the prior mean:

$$\mu_{MAP}^* = \arg \max_{\mu} \mathcal{P}(\mu; \sigma^2, \mu_0, \sigma_0^2) = \underbrace{\frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2}}_{\text{weight of the data}} \cdot \underbrace{\bar{x}}_{\substack{\text{sample mean} \\ \text{(MLE)}}} + \underbrace{\frac{\sigma^2}{n\sigma_0^2 + \sigma^2}}_{\text{weight of the prior}} \cdot \underbrace{\mu_0}_{\text{prior mean}}$$

The two weights sum to 1, so the MAP estimate always lies **between** $\bar{x}$ and μ_0 :

- $n \rightarrow \infty$ or $\sigma_0 \rightarrow \infty$ (vague prior) $\Rightarrow \mu_{MAP}^* \rightarrow \bar{x}$, i.e. MLE;
- n small or σ_0 small (confident prior) $\Rightarrow$ the estimate is pulled towards μ_0 .

By hand — one measurement $x_1 = 121$, known $\sigma = 6$, prior $\mu_0 = 120, \sigma_0 = 12$ (with three measurements the data weight would already be 0.923 — the blood-pressure example at the end):

$$\text{weights } \frac{144}{144+36} = 0.8 \text{ and } 0.2, \quad \mu_{MAP}^* = 0.8 \cdot 121 + 0.2 \cdot 120 = 120.8 \text{ mmHg}$$

Pros

- an analytic solution is usually possible when **conjugate priors** are used (exponential family distributions); otherwise numerical optimization can be used,
- easy to compute (but harder than MLE) and interpretable,
- **reduces overfitting** — the prior acts as a regularizer,
- converges to MLE with enough data.

Cons

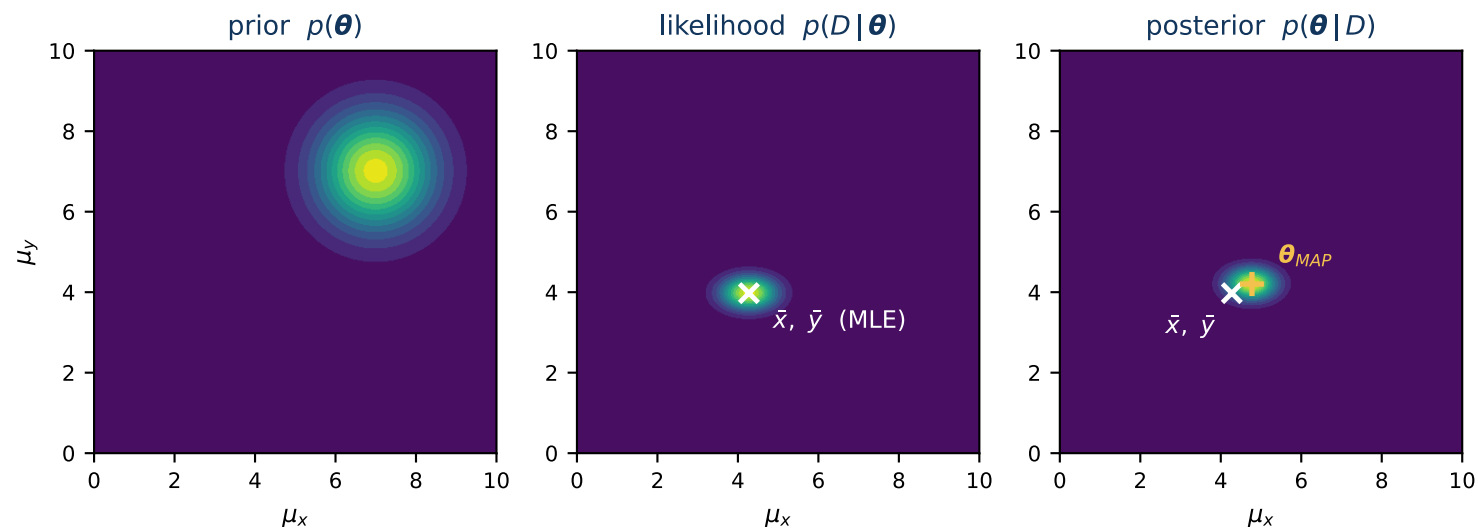
- **point estimate** — still no representation of uncertainty,
- one must assume a prior on θ .

Where does the prior come from? Population statistics, previous studies, physical limits — or an honest guess that we should then report.

5 · Bayesian inference

Stop asking for one number. Compute the whole distribution of the parameters.

The estimation method that finds the **probability distribution of the model parameters** for the observed data $D = \{x_1, \dots, x_n\}$ — a distribution of models, taking into account a prior distribution over the individual models.



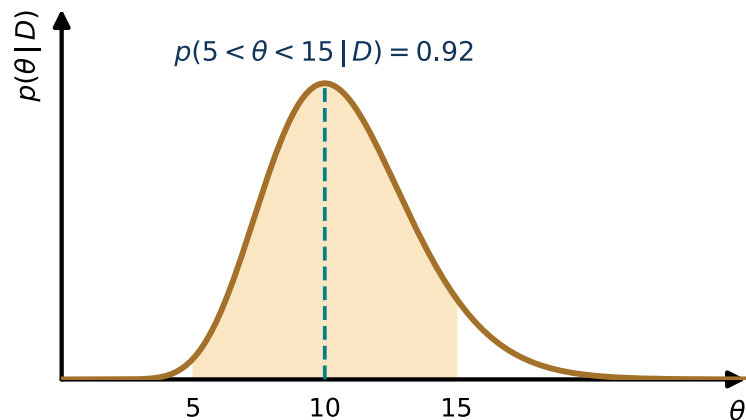
Prior $\times$ likelihood $\propto$ posterior, on a grid, for $n = 10$ measurements and a deliberately informative prior centred far from the data, at $(7, 7)$. Even so the data get the weight $\frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} = 0.82$ in μ_x and the prior only 0.18, so θ_{MAP} moves just 0.5 away from $\bar{x}$ — the posterior nearly coincides with the likelihood. MAP is the peak of the right-hand panel; Bayesian inference keeps the whole panel.

We do not get an estimate $\boldsymbol{\theta}^*$, but the whole distribution $p(\boldsymbol{\theta} \mid D)$:

$$p(\boldsymbol{\theta} \mid D) = \frac{p(D \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}{p(D)} = \frac{p(D \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}{\int_{\Theta} p(D \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) d\boldsymbol{\theta}}$$

The integral in the denominator is hard to compute — and unlike in MAP, where the `arg max` makes it irrelevant, here it **cannot be dropped**.

- **Approximation methods** — Markov Chain Monte Carlo (MCMC: Gibbs sampling, Metropolis–Hastings), Variational Inference. Very complicated; we will not use them in this course.
- **Closed-form solution** for conjugate priors — the case we will actually compute.



A **sketch** of a posterior — any skewed density with its mode at $\theta = 10$ would do. The shaded probability is the real integral of the one drawn; the blood-pressure example later gives the same kind of number from a posterior computed from measurements.

With a known distribution we get the **uncertainty** of the estimate for free.

We know not only that the best model is $\theta = 10$, but we can also compute

$$p(5 < \theta < 15 \mid D) = \int_5^{15} p(\theta \mid D) d\theta = 0.92$$

That is something no point estimate can give:

- MLE and MAP answer “*which model?*”,
- Bayesian inference answers “*which models, and how much do I believe each of them?*”

For MLE and MAP we get a single model with the optimal $\boldsymbol{\theta}^*$, and the output distribution is defined by that single model:

$$p(x_{\text{predict}} \mid \boldsymbol{\theta}^*) = p_{\boldsymbol{\theta}^*}(x_{\text{predict}})$$

For Bayesian inference we know the posterior of $\boldsymbol{\theta}$, so we can compute the **posterior predictive distribution**, which considers all models weighted by their probability:

$$p(x_{\text{predict}} \mid D) = \int p(x_{\text{predict}} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta} \mid D) d\boldsymbol{\theta}$$

It still has a closed-form solution for a conjugate prior, and it is a more honest prediction — it is **wider** than the single-model prediction, because it also contains the uncertainty about $\boldsymbol{\theta}$.

Conjugate priors — Bernoulli and Beta

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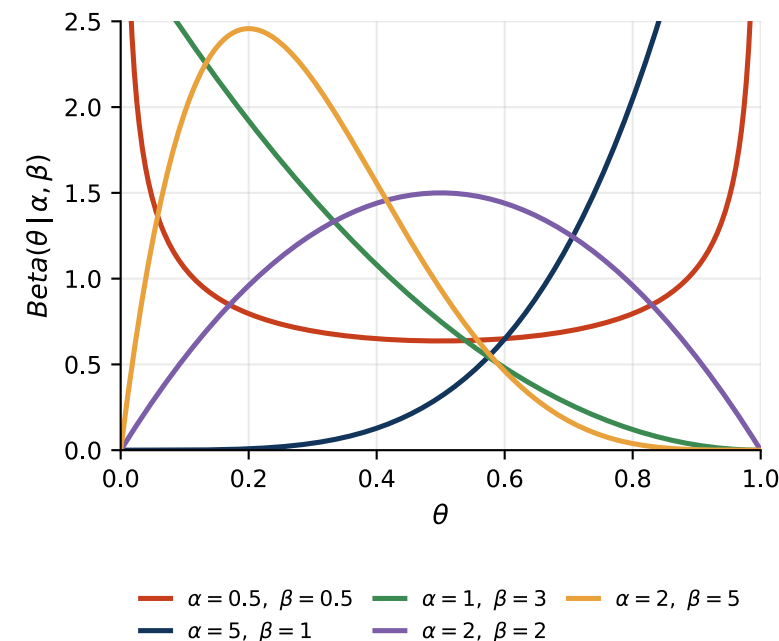
Likelihood — Bernoulli, prior — Beta, posterior — Beta:

$$\text{Beta}(\alpha', \beta') \propto \text{Bernoulli}(\theta) \text{Beta}(\alpha, \beta)$$

$$p(x \mid \theta) = \theta^x (1 - \theta)^{1-x}$$

$$\text{Beta}(\theta \mid \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

The Beta family covers everything we might believe about a probability $\theta \in [0, 1]$: flat ($\alpha = \beta = 1$), symmetric, one-sided, even U-shaped.



Bernoulli and Beta — the update

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Posterior distribution parameters — pure arithmetic:

$$\alpha' = \alpha + \sum_{i=1}^n x_i, \quad \beta' = \beta + n - \sum_{i=1}^n x_i$$

MAP estimate and posterior predictive:

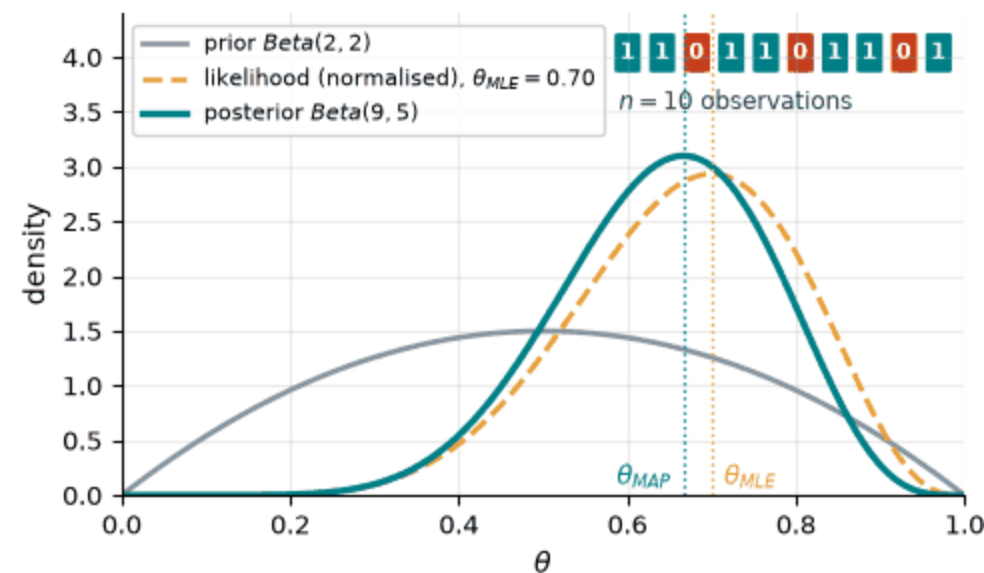
$$\theta_{MAP} = \frac{\sum_{i=1}^n x_i + \alpha - 1}{n + \alpha + \beta - 2}, \quad p(\tilde{x} = 1) = \frac{\alpha'}{\alpha' + \beta'}$$

Example — $n = 10$ trials, $\sum_i x_i = 7$ successes, prior $Beta(2, 2) \Rightarrow$ posterior $Beta(9, 5)$:

$$\theta_{MLE} = \frac{7}{10} = 0.700, \quad \theta_{MAP} = \frac{8}{12} = 0.667$$

$$p(\tilde{x} = 1) = \frac{9}{14} = 0.643$$

Three different numbers from the same data. The animation adds the observations 1, 1, 0, 1, 1, 0, 1, 1, 0, 1 one at a time — a 1 adds one to α , a 0 adds one to β . [code example — conjugate Beta prior](#)



Likelihood — normal with known variance σ^2 , prior — normal, posterior — normal:

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad \mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$$

Posterior distribution parameters:

$$\mu' = \frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \cdot \bar{x} + \frac{\sigma^2}{n\sigma_0^2 + \sigma^2} \cdot \mu_0 = \mu_{MAP}, \quad \sigma'^2 = \left(\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}\right)^{-1}$$

Posterior predictive:

$$\mathcal{N}(x \mid \mu', \sigma'^2 + \sigma^2), \quad \text{compare with } \mu_{MLE} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

The **mean** of the posterior is exactly the MAP estimate from the previous block — but now we also get σ'^2 , the uncertainty of the estimate, and $\sigma'^2 + \sigma^2$, the width of the prediction for a new measurement.

Pros

- **not a point estimate** — provides the whole probability distribution of the model parameters, so we know the model uncertainty,
- reduces overfitting,
- can provide an exact **posterior predictive** distribution,
- automatic model selection (through the marginal likelihood).

Cons

- one must assume a prior,
- exact computation can be **intractable** — the integral in the denominator.

In deep learning the integral is hopeless, so the practical compromise is MAP: weight decay is a Gaussian prior, and that is all the Bayes most networks ever see.

Example — MLE and MAP for blood pressure

Model ... a normal distribution of the measurement, $x_i \sim \mathcal{N}(\mu, \sigma^2)$ D ... a set of blood pressure measurements of one subject,
 $D = \{x_1, \dots, x_n\}$

We assume a normal distribution of the blood pressure measurement and look for the parameters of the posterior distribution $p(\mu \mid D)$. In this example σ is **known** — it is the accuracy of the sphygmomanometer — so the estimation reduces to the mean μ only.

MLE solution

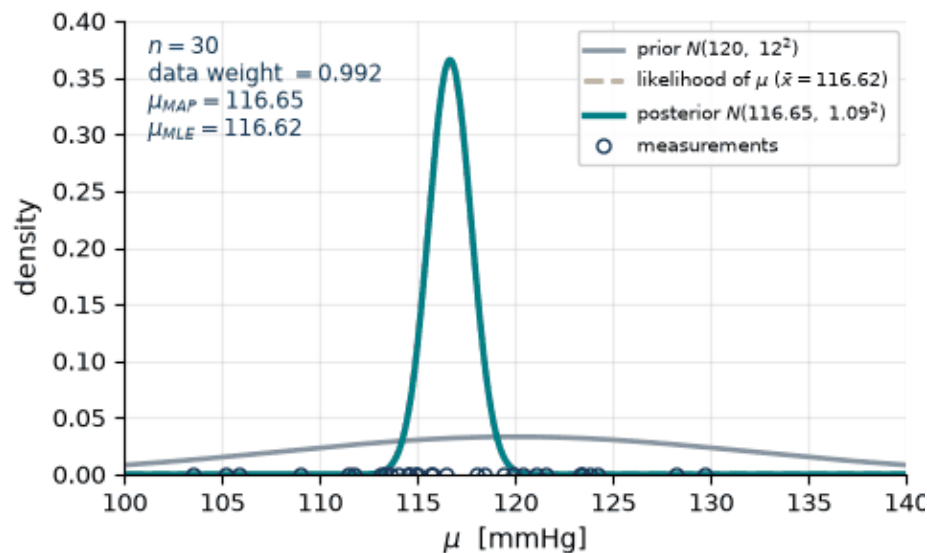
$$\mu_{MLE}^* = \arg \max_{\mu} \log p(D \mid \mu; \sigma^2) = \bar{x}$$

MAP solution — the conjugate prior is $\mathcal{N}(\mu_0, \sigma_0^2)$, where μ_0 is the mean blood pressure in the population and σ_0 its standard deviation in the population:

$$\mu_{MAP}^* = \arg \max_{\mu} \log [p(D \mid \mu, \sigma^2) \cdot p(\mu \mid \mu_0, \sigma_0^2)] = \frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \bar{x} + \frac{\sigma^2}{n\sigma_0^2 + \sigma^2} \mu_0$$

Observed ($n = 3$ measurements):

	x_1	x_2	x_3		
Observed	121	109	115		
				μ	σ
Measurement (likelihood)				—	6
Prior (population)				120	12



MLE solution

$$\mu_{MLE}^* = \bar{x} = \frac{121 + 109 + 115}{3} = 115 \text{ mmHg}$$

MAP solution

$$\mu_{MAP}^* = \frac{3 \cdot 12^2}{3 \cdot 12^2 + 6^2} \cdot 115 + \frac{6^2}{3 \cdot 12^2 + 6^2} \cdot 120 = 115.38 \text{ mmHg}$$

The weights are $\frac{432}{468} = 0.923$ for the data and $\frac{36}{468} = 0.077$ for the prior — three measurements already outweigh the population.

The animation keeps measuring the same subject (further values drawn from $\mathcal{N}(115, 6^2)$): the data weight goes $0.8 \rightarrow 0.923 \rightarrow 0.992$ for $n = 1, 3, 30$ and MAP merges with MLE — the convergence promised on the pros-and-cons slide.

Example — Bayesian inference solution

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Posterior distribution

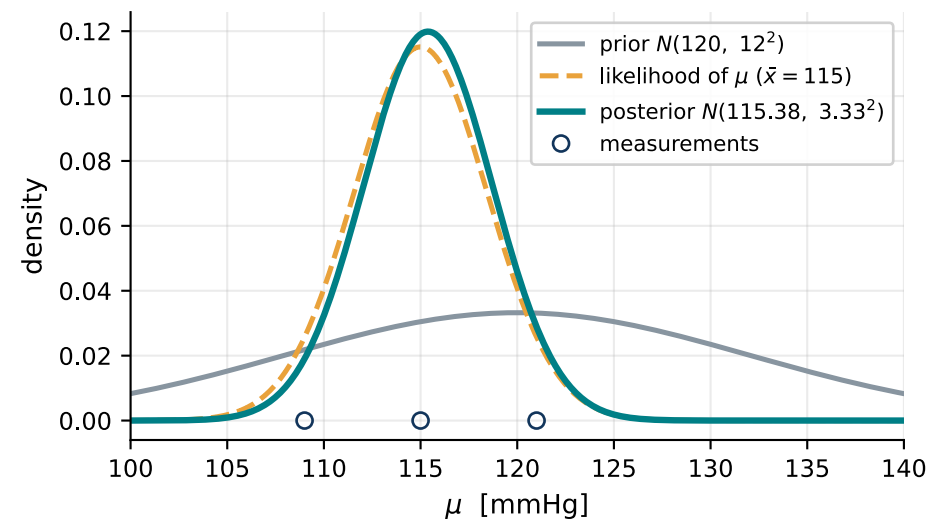
$$\mu' = \mu_{MAP}^* = 115.38 \text{ mmHg}$$

$$\sigma'^2 = \left(\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2} \right)^{-1} = \left(\frac{1}{12^2} + \frac{3}{6^2} \right)^{-1} = 11.08 \Rightarrow \sigma' = 3.33$$

Posterior predictive — for the **next** measurement of the same subject:

$$\mathcal{N}(x \mid \mu', \sigma'^2 + \sigma^2) = \mathcal{N}(x \mid 115.38, 11.08 + 6^2 = 47.08)$$

i.e. a standard deviation of 6.86 mmHg.



MLE and MAP are easy to compute and intuitively interpretable, often with an analytical solution; without one, numerical optimization works too. They provide only a **single estimate** of the parameters, without uncertainty.

Maximum likelihood

- maximizes $p(D \mid \theta)$ — the model that best generates the observed data,
- the estimation is based **only on the observed data**,
- prone to overfitting.

Maximum a posteriori probability

- maximizes $p(\theta \mid D)$ — the most probable model given the observed data,
- requires a prior $p(\theta)$, which favours some distributions over others,
- the prior acts as a **regularization** and reduces overfitting.

Bayesian inference is hard to compute (except for special cases such as conjugate priors). In exchange it provides the **full probability distribution** of the model parameters, and the **posterior predictive distribution** — a mixture of the predictions of all models weighted by their probability. It is similar to MAP, but returns a distribution instead of a point.

	what is maximized / computed	needs a prior	output	overfitting
MLE	$p(D \mid \theta)$	no	one θ^*	yes
MAP	$p(D \mid \theta) p(\theta)$	yes	one θ^*	regularized
Bayesian inference	$p(\theta \mid D)$ whole	yes	a distribution over θ	regularized

Blood pressure example, all three from the same three measurements: $\mu_{MLE} = 115.00$, $\mu_{MAP} = 115.38$, posterior $\mathcal{N}(115.38, 3.33^2)$, prediction $\mathcal{N}(115.38, 6.86^2)$.

Done today — Probabilistic models 1

- probability introduction and Bayes' rule,
- parameter estimation: maximum likelihood, maximum a-posteriori, Bayesian inference.

Next lecture — Probabilistic models 2

- naive Bayes classifier,
- Gaussian mixture model and the EM algorithm,
- logistic regression,
- and where loss functions come from: L2 loss is MLE under Gaussian noise, cross-entropy is MLE under a Bernoulli likelihood.

6 · Recap

Three sets of questions: probability, estimation, and the three methods.

- What is the difference between a **PMF** and a **PDF**? Why can a density exceed 1, and what does $P(X = x) = 0$ mean for a continuous variable?
- Write the **chain rule** and the **law of total probability**. Which of them needs independence?
- A contingency table has counts. How do you get the **joint**, the **marginal** and the **conditional** distribution from it?
- Derive **Bayes' rule** from the symmetry of the joint distribution. Name all four terms.
- A test has sensitivity and specificity 98 %. Why does a positive result mean 84 % at a prevalence of 10 % and only 33 % at 1 %? Which quantity from the second lecture is that?
- On the shapes-and-colours example: why is $p(b \mid s) \neq p(s \mid b)$ while $p(b, s) = p(s, b)$?

- Define **prior**, **likelihood**, **posterior** and **marginal likelihood**. Which of them is not a distribution over θ ?
- Why does the i.i.d. assumption turn the likelihood into a **product**? Why do we take its logarithm?
- Derive $\hat{\mu}_{MLE} = \bar{x}$ for a normal distribution. Which term of the log-likelihood does not contain μ at all?
- Why is $\hat{\sigma}_{MLE}^2$ **biased**, and how big is the bias for $n = 50$?
- Name three weaknesses of MLE and match each with one of the two sketches of a likelihood function.
- What is the **black swan paradox** and what does it say about point estimates?

- Write θ_{MAP}^* . Which two terms may be dropped inside the $\arg \max$, and under what condition each?
- What is a **conjugate prior** and why do we care? Give the pair for a Bernoulli likelihood and for a normal likelihood.
- μ_{MAP}^* is a convex combination of what? What happens to the weights when $n \rightarrow \infty$, and when $\sigma_0 \rightarrow \infty$?
- Prior $Beta(2, 2)$, 7 successes out of 10: compute the posterior, θ_{MLE} , θ_{MAP} and the posterior predictive. Why are they three different numbers?
- Why can Bayesian inference **not** drop $p(D)$, while MAP can?
- What is the **posterior predictive** distribution, and why is it wider than the prediction of the single best model?
- Blood pressure example: the posterior is $\mathcal{N}(115.38, 3.33^2)$. Where did 3.33 come from and why is it smaller than both σ and σ_0 ?